

**“How Concern for Animals Relates to Caregiving Behaviours, Personality Traits, and
Narcissism in Young Adults”**

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Declaration

This project is submitted in partial fulfilment for the Degree of Bachelor of Science in Psychology and Disruptive Technologies at Dublin City University, 13th March, 2026, and, except where otherwise acknowledged, this is my own unaided work.

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Abstract

Human-to-animal interactions have been associated with broader moral and interpersonal functioning. However, important gaps remain in the literature and the relationship between empathy towards animals, and prosocial behaviour, personality traits such as agreeableness and conscientiousness, and narcissism remains underdeveloped. These traits are understood to be central to social functioning, highlighting the importance of determining if empathy towards animals could act as a potential indicator of personality styles and social behaviours. Therefore, the current study aimed to explore the predictive nature of empathy towards animals and its relationship with prosocial behavior, agreeableness, conscientiousness and narcissism in young adults. An online survey was administered via Qualtrics and advertised through social media. The final sample included 112 participants aged 18–25 ($M = 22$, $SD = 1.66$). Measures included The Animal Empathy Scale (AES), The Prosocialness Scale for Adults (PSA), the Agreeableness and Conscientiousness subscales of the NEO Five-Factor Inventory, and the Narcissistic Personality Inventory (NPI). Four simple linear regressions were conducted, with Empathy for Animals serving as the independent variable across all models. The findings indicated that higher levels of empathy towards animals significantly predicted higher levels of prosocial behaviour, agreeableness and conscientiousness. Conversely, lower levels of empathy towards animals significantly predicted higher levels of narcissism. In conclusion, the positive and negative relationships observed suggests that empathy towards animals reflects broader interpersonal tendencies. However, when interpreting these results, it is important to consider the generalizability of the sample and social desirability responding.

Introduction

Empathy for animals has increasingly been recognised as an important aspect of social and moral functioning, linked to broader personality traits and prosocial behaviour (O'Haire et al., 2014; Signal & Taylor, 2007). However, as with many psychological constructs, empathy is strongly influenced by individual differences. These individual differences can be observed not only in human-to-human relationships but also in human-to-animal interactions. Growing research suggests that concern for animals reflects broader empathic orientation, positioning human-to-animal directed behaviours as indicators of interpersonal functioning. However, several areas of this research field remain unexplored. The present study therefore aims to directly examine the predictive relationship between concern for animals and prosocial behaviour, adaptive personality traits, and narcissism.

Empathy

Among the many perspectives from which researchers attempt to define empathy, Eisenberg (2000) claims it to be the recognition of an emotional state in others, which then prompts an emotional response similar to the emotional state of another. Empathy consists of two components, affective empathy, the ability to feel and share another's emotional or cognitive state, and cognitive empathy, the recognition and understanding of another's emotional or cognitive state (Blair, 2005; Chakrabarti & Baron-Cohen, 2006). Successful social interaction depends on the integration of both components, as individuals must accurately interpret others' emotions while responding in an emotionally appropriate manner. These components co-occur in everyday interactions and can be difficult to separate in practice (Baron-Cohen & Wheelwright, 2004). However, neuroimaging suggests they are distinct processes (Bzdok et al., 2012), with some individuals showing stronger tendencies in one component than the other.

Empathy is recognised as an essential component for higher social functioning and has been associated with increased emotional sensitivity (Baron-Cohen & Wheelwright, 2004; Findlay et al., 2006). Conversely, a lack of empathy is seen to predict antisocial behaviour, leading to struggles in forming and maintaining stable social relations (Blair, 2010; Eisenberg, 2000). Given this central role of empathy in regulating social behaviour, exploring how the construct extends beyond human-to-human interactions offers a valuable extension to the research.

Empathy towards Animals

Empathy towards animals refers to the capacity to recognise, emotionally respond to, and share in the experiences and welfare of non-human animals (Paul, 2000). Because animals cannot verbally express mental states, empathy for animals relies heavily on the affective component of empathy. This is emphasised by Preston & De Waal (2003) who argue empathy is strongly reliant on direct perception of distress, explaining that perceiving distress in another activates the same neural and emotional systems that would be active if the observer were distressed.

There is increasing support for the idea that human attitudes shown towards animals may also be reflected in their human-to-human interactions. Experimental research has demonstrated that individuals often experience empathy towards animals that is equal to empathy towards humans. Upon reading written scenarios describing either a human or an animal victim of abuse, participants reported their empathic responses. Adult women reported the same levels of empathy for a child and a puppy, and both men and women expressed the same levels of empathy for a baby as for a puppy (Angantyr et al., 2015). Similarly, concern for animal welfare

has been associated with higher levels of human-directed empathy, with those expressing more protective attitudes towards animals also reporting greater empathic concern for others (Signal & Taylor, 2007). Conversely, lack of concern for animals and animal cruelty has been significantly linked to antisocial personalities (Gleyzer et al., 2002).

Taken together, this evidence suggests that empathy towards animals is supported by the same emotional processes involved in human empathy and is closely linked to broader interpersonal functioning.

Prosocial Behaviour and Empathy

Batson and Powell (2003) define prosocial behaviour as a broad range of actions intended to benefit another individual, other than one self. Prosociality can come in various forms such as helping, comforting, taking care of the sick or sharing. It plays a central role in the formation and maintenance of social relationships, encouraging trust, cooperation, and attachment within groups (Baldassari & Grossman 2013).

Prosocial behaviour has been explained through a number of theoretical perspectives, which differ in the extent to which helping is viewed as motivated by self-interest or emotional processes. Some theorists propose that helping is motivated by expectations of reciprocity (Trivers, 1971). However, other research outlines not all helping behaviours are motivated by self-interest. Empathic concern has been identified as a key driver of genuine other-oriented helping. The empathy–altruism hypothesis (Batson, 1982) proposes that when people feel empathic concern for someone in need, they are motivated to help for solely altruistic reasons, (i.e., the primary benefit is experienced by the other person, rather than themselves).

Across multiple studies, researchers tested alternative egoistic explanations for this effect, including the negative state relief hypothesis, when individuals help others to reduce their own temporary feelings or sadness or guilt (Cialdini et al., 1987), or empathy-specific motivations, whereby individuals help seeking positive self-evaluations or to avoid feelings such as guilt, shame, or social disapproval (Batson et al., 1988). However, when opportunities for mood improvement, external rewards, and social consequences were removed, individuals high in empathy continued to help more than those low in empathy, supporting empathy as a genuine driver of prosocial behaviour (Batson et al., 1988). Within this framework, prosocial and caregiving behaviours can be understood as behavioural outcomes of empathic concern.

The need to belong theory proposes that humans are inherently motivated to form and maintain close relationships, with prosocial behaviours helping to sustain these bonds (Baumeister & Leary, 1995). Empathy plays a central role in this process, as sensitivity to others' distress promotes concern and motivation to help (Eisenberg & Miller, 1987). In turn, these supportive social relations are associated with increased psychological wellbeing, further highlighting the importance of empathy in interpersonal functioning (Holt-Lunstad et al., 2010).

Prosociality is seen to extend beyond human interactions and into experiences with animals. A meta-analysis revealed a positive association between human-animal interactions and prosocial behaviours, supporting the importance of this relationship in driving prosocial and caregiving behaviours (Chen et al., 2023). With this reasoning, empathy towards animals may also be conceptualized as a potential predictor of prosociality.

Personality Traits and Empathy

Allport (1937) described personality as a dynamic organization within the individual that determines their unique adjustments to the environment. Extending this understanding of personality, Costa & McCrea (1999) described personality traits as the relatively stable patterns of thoughts, feelings, and behaviours an individual exhibits across different life experiences.

Personality traits shape how one perceives, interprets, and responds to others' emotional states, which is central to understanding empathic concern. The five-factor model, developed by Costa & McCrea (1992) describes personality into five stable dimensions, including Conscientiousness, Openness to Experience, Extraversion, Neuroticism, and Agreeableness. Among these, agreeableness has been most consistently linked to empathy and prosocial behaviour (Tariq & Naqvi, 2020; Caprara et al., 2010). Extending this, cross-cultural research found that agreeableness and conscientiousness were the most prominent personality predictors of empathy in university students (Melchers et al., 2016).

Empirical research has shown that agreeableness is associated with greater emotional sensitivity, altruism, and interpersonal warmth, all of which are central to empathic concern (Costa & McCrae, 2008; Robinson et al., 2024). As a result, individuals high in agreeableness tend to engage more frequently in caregiving and helping behaviours, suggesting a strong theoretical link between this trait and empathy towards both humans and animals (Graziano et al., 2007).

Conscientiousness, while less directly emotional, is characterised by responsibility, reliability, and self-regulation, and has been associated with moral behaviour (Roberts et al.,

2014). Individuals high in conscientiousness are more likely to act consistently in ways that consider others' needs, as their sense of duty promotes attentive and responsible caregiving.

Airagnes et al. (2021) examined the relationship between personality and cognitive empathy, finding that conscientiousness was positively associated with baseline levels of cognitive empathy in medical students. In this context, empathy was defined as the capacity to understand patients' experiences and perspectives, suggesting that conscientiousness may relate more to deliberative perspective-taking than to affective emotional responsiveness.

Recent research on pet attachment and personality found conscientious individuals to be strongly correlated with stronger emotional bonds with pets (Wells & Treacy, 2024), suggesting this trait may promote empathy through sustained care and responsiveness.

However, these traits may not be entirely stable, as research suggests their expression can change in response to life experiences. For instance, depressive symptoms have been associated with decreases in agreeableness, suggesting that empathic and prosocial tendencies may also change accordingly (Hakulinen et al., 2015). Furthermore, highly conscientious individuals have been shown to experience declines in wellbeing following prolonged unemployment, implying that disrupted goal-directed roles may heighten distress (Boyce et al., 2010).

Narcissism and Empathic Deficits

Narcissism is considered a maladaptive dimension of personality and is modernly characterised by grandiosity, entitlement, and an excessive focus on the self (Miller et al., 2021). Narcissistic Traits have consistently been associated with empathic deficits. Findings show this

is expressed even towards loved ones, and that narcissistic individuals don't just care less, they also react emotionally out of sync with others (Burgmer et al., 2019).

Interestingly, narcissistic individuals mainly show deficits in affective empathy rather than cognitive empathy (Ritter et al., 2011), meaning understanding another's perspective remains intact when useful, for example, when it serves a self-relevant goal such as praise. A narcissistic individual may recognise when someone is distressed but lack the emotional motivation for compassionate care. This has potential implications for the ability to care for animals as effective care requires sensitivity to another's needs and emotional cues (Eisenberg & Miller, 1987; Preston & De Waal, 2003). Studies examining attitudes towards animals indicate that individuals with higher narcissistic traits tend to hold more negative views of animals, which is associated with lower levels of animal-centered empathy (Kavanagh et al., 2013; Ellingsen et al., 2010). Collectively, this research suggests that narcissism may undermine empathy towards animals through impaired emotional engagement and resonance, rather than a lack of understanding, in turn limiting the capacity for concern and empathy for animals.

While much of the literature suggests stable associations between empathy, caregiving behaviours, and personality traits, the general perception may be less consistent than it appears and important gaps remain.

First, much of the literature has focused on pet ownership, attachment, or attitudes toward animal welfare, rather than measuring empathy towards animals as a stable individual difference. Second, although existing findings suggest links between animal-related orientations and human social functioning, research seems to be fragmented and few studies have directly tested whether empathy towards animals predicts everyday interpersonal tendencies towards other humans.

Finally, relatively little work can be found examining these behaviours in young adulthood, a key developmental period for personality, moral values, and social responsibility (Arnett, 2000).

The present study aims to address these gaps by directly examining the relationship between empathy towards animals, prosocial behaviour, personality traits, and narcissism in young adults. Thus, the following hypotheses were proposed:

H1: Greater concern and empathy for animals (higher AES scores) will be associated with higher levels of caregiving and prosocial behaviours (higher PSA scores).

H2: Concern and empathy for animals will significantly predict personality traits, particularly agreeableness and conscientiousness.

H3: Greater concern and empathy for animals will be negatively associated with narcissistic traits, such that higher empathy towards animals will predict lower levels of narcissism.

Methodology

Design

This study uses a quantitative, cross-sectional, correlational survey design to examine the relationships between empathy for animals, prosocial behaviours, adaptive personality traits, and narcissism in young adults.

Participants

157 participants were recruited for the study in total. 112 participants met the inclusion criteria and finished at least 85% of the survey and therefore retained for analysis. All participants were aged between 18 and 25 years and included 39 males, 69 females, 3 non-binary

participants, and 1 participant preferred not to say. All participants were recruited via social media advertisements on Instagram and LinkedIn.

Due to a survey design error, the first 59 participants were asked to confirm that they were aged between 18–25, rather than entering their exact age. This was later corrected to request a specific age in years. For participants who provided an exact age ($n = 53$), the average age was 22 years ($SD = 1.66$). All participants were confirmed to fall within the inclusion age range of 18–25.

Materials

Recruitment materials included use of social media platforms such as LinkedIn and Instagram, a recruitment poster (See appendix A) with a brief description of the study, eligibility criteria, and a QR code linking directly to the anonymous online survey.

Measures

Given the inclusion of four psychological measures within the survey, each scale was slightly reduced in order to minimize participant burden and reduce the risk of survey fatigue, as these are identified as key limitations of questionnaire-based research, resulting in reduced data quality (Ghafourifard, 2024). The critical considerations for reductions, which were applied for the present study, is that retained items should stay aligned with the study's aims and maintain acceptable psychometric performance (Cruchinho et al., 2024). All items included for the present study and their corresponding psychometric properties, are provided in Appendices B to F.

Animal Empathy Scale (AES; Paul, 2000)

The AES is used to assess concern and emotional responsiveness towards animals. In the present study, empathy towards animals was assessed using 16 selected items from the original 22-item AES. Items that were context dependent (e.g., “I enjoy feeding scraps of food to the

birds” or “Animals deserve to be told off when they’re not behaving properly”) were excluded, as they may reflect situational or cultural experiences. The retained items aligned closely with the study’s focus on generalised empathy toward animals. Items were rated on a 5-point Likert scale ranging from 1 (Strongly disagree) to 5 (Strongly agree). Negatively worded items were reverse scored prior to analysis. Higher total scores indicated greater empathy toward animals. Internal consistency was excellent ($\alpha = .93$).

Prosocialness Scale for Adults (PSA; Caprara et al., 2005)

Prosocial behaviour was assessed using 14 selected items from the original 16 item PSA. Items measured helping, sharing, and comforting behaviours towards others. Responses were rated on a 5-point Likert scale ranging from 1 (Never) to 5 (Always). Higher total scores indicated greater prosocial behaviour. Internal consistency was excellent ($\alpha = .93$).

NEO Five-Factor Inventory (NEO-FFI; Costa & McCrae, 2008)

Agreeableness and Conscientiousness were assessed using 6 selected items from each corresponding 12-item NEO-FFI subscale. Responses were rated on a 5-point Likert scale ranging from 0 (Strongly disagree) to 4 (Strongly agree). Negatively worded items were reverse scored prior to analysis. Internal consistency was acceptable for Agreeableness ($\alpha = .65$) and Conscientiousness ($\alpha = .72$).

Narcissistic Personality Inventory (NPI-40; Raskin & Hall, 1979)

Narcissistic traits were assessed using 11 selected items from the NPI-40. For each pair, participants selected the statement that best described them (e.g., “I like to be the center of attention” (narcissistic) or “I prefer to blend in with the crowd” (non-narcissistic)). Narcissistic responses were coded as 1 and non-narcissistic responses as 0 and each narcissistic response was summed to create a total narcissism score. Internal consistency was acceptable ($\alpha = .71$).

Procedure

Participants accessed the survey via Qualtrics. Upon entry, they were presented with a Participant Information Sheet outlining the study's aims, voluntary participation, confidentiality, and withdrawal rights. Participants then provided informed consent before continuing.

Participants first completed demographic questions, followed by the AES items, PSA items, NEO-FFI subscale items, and NPI items. The survey took approximately 9 minutes to complete. Upon submission, participants were presented with a debrief statement.

Data Analysis

Survey responses were exported from Qualtrics and imported to IBM SPSS (Version 30). Prior to analysis, missing data and incomplete responses were checked for. Participants who completed less than 85% of the questionnaire were removed from the dataset. Negatively worded items in the AES and NEO-FFI Subscales were reverse scored. Each scale was coded numerically so that higher total scores reflected higher levels of the construct being measured. The NPI was coded so that narcissistic responses were scored as 1 and non-narcissistic responses were scored as 0, then each narcissistic response was summed to create a continuous total score of narcissism levels.

Descriptive statistics were calculated for all study variables. Internal consistency and reliability was assessed using Cronbach's alpha coefficients. Assumptions of linear regression were assessed and met. Statistical significance was set at $p < .05$.

Pearson's correlations were conducted between all variables. As each hypothesis concerned a distinct outcome, and in order to examine the predictive relationship of empathy towards animals on each construct independently, a series of four simple linear regressions were carried out. In each model, empathy towards animals served as the independent variable.

Separate regression analyses were conducted with (1) prosocial behaviour, (2) agreeableness, (3) conscientiousness, and (4) narcissism as dependent variables.

Ethical Considerations

Ethical approval was obtained for this study by Dublin City University's School of Psychology Ethics committee (PEC) on 02/01/2026 (approval code: DCUPEC_2025_135) (See appendix G).

The study was classified as low risk. Participation was voluntary and anonymous, and no identifying information was collected. Participants were informed of their right to withdraw at any time.

Although minimal psychological discomfort was anticipated due to personality related questions, participants were advised they could skip any question or withdraw at any time. A debrief page with reassurance and relevant support information was provided at the end.

Results

Descriptive Statistics

The final sample retained for analysis ($n = 112$) included 39 males (34.8%), 69 females (61.6%), 3 non-binary (3.7%), and 1 participant preferred not to say (0.9%). All participants were aged 18-25. For participants who answered the corrected question, the average age was 22 ($n = 53$, $SD = 1.66$). Internal Consistency across scales ranged from acceptable to excellent (.65 - .93). Mean scores from all respondents indicated relatively high levels of empathy for animals and prosocialness, with moderately high scores observed on the NEO-FFI subscales and low

levels of narcissism across the sample. Descriptive statistics for all study variables are presented in Table 1.

Table 1

Descriptive Statistics and Internal Consistency of Study Variables (N=112)

	Minimum	Maximum	M	SD	α
Empathy towards Animals	16	80	63.29	11.42	.93
Prosocialness	27	70	56.78	8.62	.93
Agreeableness	11	29	22.14	3.39	.65
Conscientiousness	13	30	21.67	3.45	.72
Narcissism	0	11	2.69	2.2	.71

Data Preparation

Prior to analysis, univariate outliers were checked for using standardized Z-scores. Cases with z-scores exceeding ± 3 were identified as potential outliers. Across variables, standardized scores ranged from -4.14 to 3.77 , with two participants accounting for the majority of extreme scores across scales. However, inspection of data indicated these scores were possible, and were retained for analysis.

Multivariate outliers were assessed using Mahalanobis distance, with a critical chi-square value of 10.83 ($df = 1$, $p < .001$). A small number of cases exceeded this threshold across models. However, Cook's Distance values remained below 1.00 in all analyses (ranged from $.20$ to $.40$), indicating no case had an excessive influence on the model's estimates.

Regression models were examined with and without extreme cases. Removal did not substantially affect regression coefficients, R^2 values, or statistical significance. Therefore, all cases were retained in final analyses. Collinearity diagnostics indicated no multicollinearity concerns ($VIF = 1.00$). Durbin–Watson statistics ranged between 1.82 to 2.07 across models.

Normality

The assumptions of linear regression were sufficiently met through inspection of residuals using standardized residual statistics, histograms, and P–P plots. Standardized residuals had a mean of .00 and a standard deviation of .99, consistent with expectations under normality. Visual inspection of the histogram indicated that the standardized residuals were normally distributed, with a bell-shaped pattern that centered around zero. The P–P plot showed that values closely followed the diagonal line. Furthermore, scatterplots of predicted standardized values against standardized residuals showed a random dispersion of points.

Pearson's Correlation

Pearsons's correlation was conducted to initially examine associations between empathy towards animals, prosocial behavior, personality traits, and narcissism (See Table 3). Higher empathy for animals was strongly positively correlated with Prosocialness ($r = .70$, $n = 109$, $p < .001$, 95% CI [.59, .78]) and Agreeableness ($r = .55$, $n = 110$, $p < .001$, 95% CI [.42, .68]), both significant at a .01 level. A small positive correlation was seen between empathy for animals and conscientiousness ($r = .19$, $n = 110$, $p = .05$, 95% CI [.003, .364]), with significance at a .05 level. Empathy for animals was negatively correlated with narcissism ($r = -.52$, $n = 108$, $p < .001$, 95% CI [-.646, -.368]), at a significance level of .01.

Table 2

Pearson's Correlation between Empathy for Animals and Dependent Variables

	Empathy for Animals	Prosocialness	Agreeableness	Conscientiousness	Narcissism
Empathy for Animals	1				
Prosocialness	.70**	1			
Agreeableness	.57**	.58**	1		
Conscientiousness	.19*	.36**	.16	1	
Narcissism	-.52**	-.52**	-.54**	-.14	1

Note: $p < .05^*$, $p < .01^{**}$. N ranged from 108–110 due to pairwise deletion.

Regression Analyses

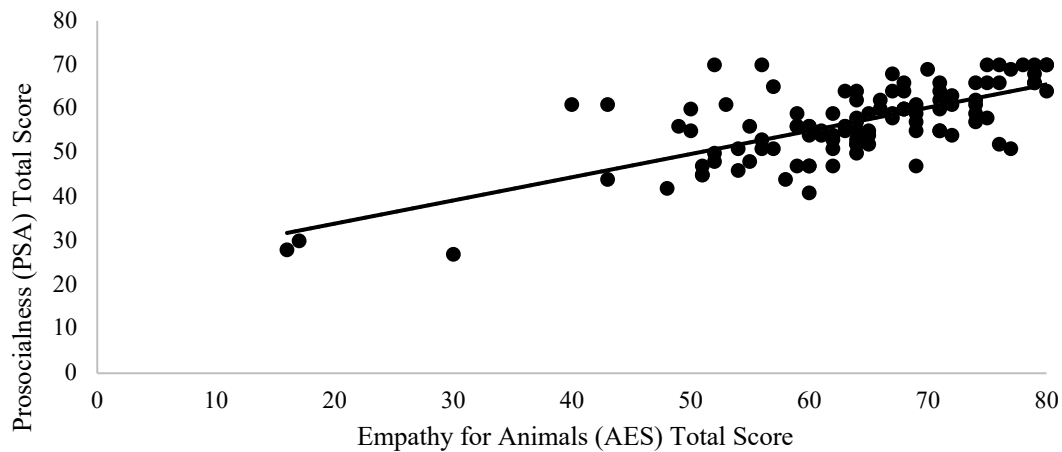
Four simple linear regression analyses were carried out in order to explore the predictive nature of Empathy for Animals. Table 3 presents the results of each model.

Model 1: Empathy for Animals and Prosocialness

The first model was statistically significant, $F(1, 107) = 101.72$, $p < .001$, with empathy for animals accounting for 48.7% of the variance in prosocial behavior ($R^2 = .49$, adjusted $R^2 = .48$). In support of Hypothesis 1, empathy for animals had a statistically significant positive predictive effect on prosocial behaviour. This relationship can be seen in Figure 1.

Figure 1

Relationship Between Empathy for Animals and Prosocial Behaviour

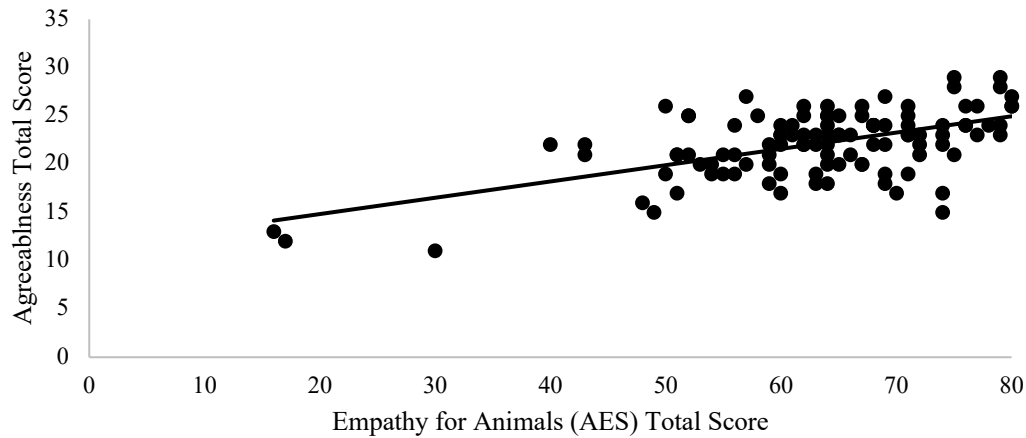


Model 2: Empathy for Animals and Agreeableness

The second linear regression analysis examined whether empathy for animals predicted Agreeableness. The model was also statistically significant, $F(1, 108) = 50.98, p < .001$, explaining 32% of the variance in agreeableness ($R^2 = .32$, adjusted $R^2 = .31$). This result supports Hypothesis 2, indicating empathy for animals is a significantly positive predictor of agreeableness. This relationship can be seen in figure 2.

Figure 2

Relationship Between Empathy for Animals and Agreeableness

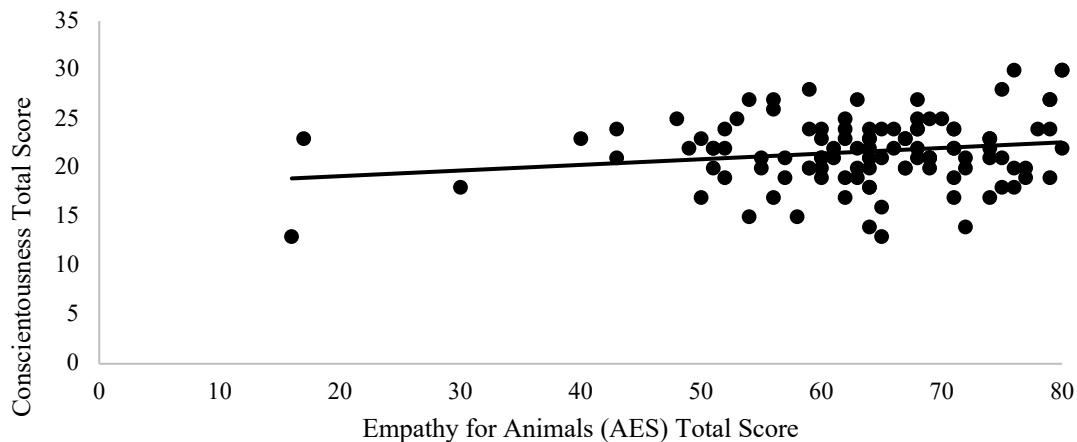


Model 3: Empathy for Animals and Conscientiousness

The third linear regression analysis examined whether empathy for animals predicted conscientiousness. The model was statistically significant, $F(1, 108) = 4.05, p < .05$. However, the effect size was small, with empathy for animals explaining 3.6% of the variance in conscientiousness ($R^2 = .04$, adjusted $R^2 = .03$). This relationship can be seen in Figure 3

Figure 3

Relationship Between Empathy for Animals and Conscientiousness



Model 4: Empathy for Animals and Narcissism

The final linear regression analysis examined whether empathy for animals predicted narcissistic tendencies. The model was statistically significant, $F(1, 106) = 39.42, p < .001$, with empathy for animals explaining 27.1% of the variance in narcissistic behaviour ($R^2 = .27$, adjusted $R^2 = .26$). As hypothesized in hypothesis 3, empathy for animals was a significantly negative predictor of narcissism. This relationship can be seen in Figure 4

Figure 4

Relationship Between Empathy for Animals and Narcissism

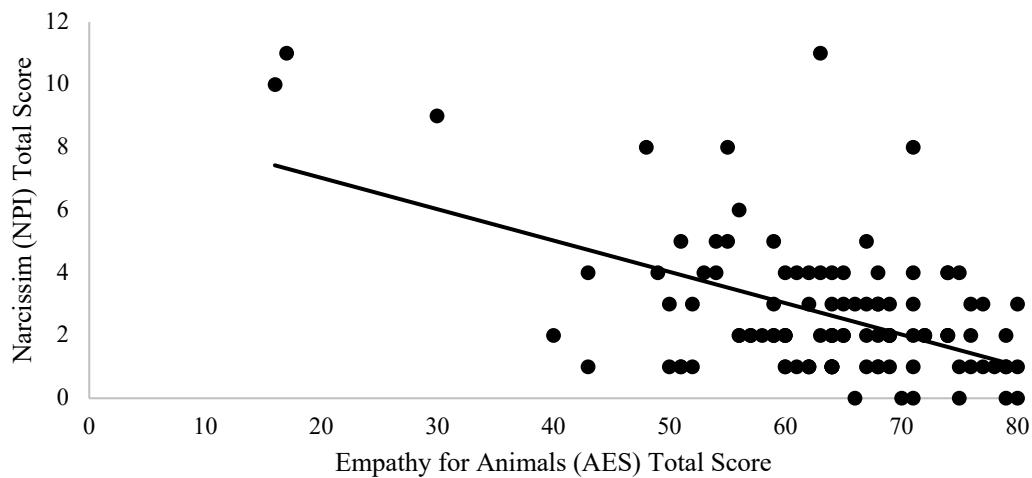


Table 3

Results of Simple Linear Regression Analysis Predicting variables from Empathy towards Animals

Outcome Variable	B	SE B	β	T	P	R ²	95% CI
Prosocialness	.53	.05	.69	10.09	<.001	.49	[0.42, 0.63]
Agreeableness	.17	.02	.57	7.14	<.001	.32	[0.12, 0.21]
Conscientiousness	.06	.03	.19	2.0	<.05	.04	[.00, .11]
Narcissism	-.10	.02	-.52	-6.28	<.001	.27	[-.13, -.07]

Discussion

The present study examined the predictive nature of empathy for animals in prosocial behaviour, agreeableness, conscientiousness and narcissism in young adults. All hypotheses proposed were statistically supported by the findings, with higher levels of empathy and concern for animals significantly predicting higher levels of prosocial behaviour and agreeableness, and lower levels of narcissism. However, it only weakly significantly predicted conscientiousness, suggesting that concern for animals may be more closely aligned with emotional responsiveness in human relationships. The literature to date has conceptualised empathy for animals as a reflection of broader empathic and moral orientation (Signal & Taylor, 2007). The present study

extended this perspective by directly examining whether concern for animals acts as an inherent indicator of prosocial tendencies and interpersonal styles in young adults.

Earlier theorists such as Batson (1982) proposed that empathic concern motivates individuals to help solely for altruistic reasons, through his empathy-altruism hypothesis. Extending this perspective, more contemporary research suggests that empathy and concern for animals can mirror the emotional processes involved in human-directed empathy (Angantyr et al., 2015). The current findings integrate both these perspectives and further support the idea that empathy for animals is a potential predictor of an individuals' helping behaviours.

Because empathy toward animals relies on affective responsiveness (Preston & De Waal, 2003), those who are emotionally attuned to animals may also be more sensitive to vulnerability cues more generally. In turn, this empathic sensitivity may to be seen to predict caregiving, compassion and prosociality. The current findings further emphasise those obtained by Eisenberg & Miller (1987), who demonstrated a strong association between empathy and helping behaviour.

In partial support of our second hypothesis, a significant positive relationship was observed between empathy towards animals and agreeableness. This finding is consistent with previous research that suggests agreeableness is closely associated with empathic concern and emotional sensitivity (Caprara et al., 2010; Costa & McCrae, 2008; Tariq & Naqvi, 2020). Given that empathy toward animals involves sensitivity to vulnerability and distress in non-human beings (Paul, 2000), and that agreeableness has consistently been associated with compassion and emotional responsiveness (Robinson et al., 2024), it is understandable that individuals high in agreeableness would also report stronger concern and understanding towards animals.

Similar to the findings of Graziano et al. (2007), which identified agreeableness as a predictor of caregiving and helping behaviours, agreeableness in the present study was not only positively associated with empathy towards animals but also demonstrated a strong correlation with prosocialness. Although this relationship was not a primary focus of the study and was simply an unanticipated secondary finding, it is consistent with prior research linking agreeableness to prosocial motivations. Importantly, whereas Graziano and colleagues (2007) focused on interpersonal helping within human contexts, the current findings suggest that this behaviour may extend beyond human-human interactions to include empathic concern towards animals.

The weak relationship observed between empathy for animals and conscientiousness provided limited support for this aspect of hypothesis 2. However, the findings were statistically significant, and a small positive relationship was observed, suggesting that concern for animals plays a comparatively weaker role in predicting conscientiousness.

This finding is not entirely consistent with existing literature. Conscientiousness is characterised by responsibility, reliability, and self-regulation, and has been consistently associated with moral behaviour (Roberts et al., 2014). Previous research indicates that conscientiousness predicts both cognitive and affective components of empathy (Melchers et al., 2016; Airagnes et al., 2021). Furthermore, Wells and Treacy (2024) reported a positive association between conscientiousness and stronger pet attachment. On this basis, it was expected that individuals who are guided by a sense of duty and morals would also report greater concern for animal welfare. Moreover, individuals who form strong emotional bonds with their pets would be expected to also demonstrate higher levels of empathy towards animals more

generally. However, the weak relationship suggests that this form of empathic concern for animals may not act in the same way.

As predicted in our third hypothesis, empathy towards animals was found to be a significantly negative predictor of narcissism. These findings suggest that individuals who demonstrate greater emotional sensitivity towards animals may be less characterised by self-focused tendencies. Consistent with contemporary perceptions of narcissism being characterised by empathic deficits and reduced emotional attunement to others (Burgmer et al., 2019; Miller et al., 2021), the current findings may build on this to provide further insight into how this extends into human-animal interactions. In support of this pattern, Ellingsen et al. (2010) showed that lower empathy was associated with reduced concern for animal welfare, while Kavanagh et al. (2013) then went on to find that narcissistic traits were linked to greater acceptance of animal cruelty. The present findings extend this research by suggesting that empathic concern towards animals more generally, may also function as an inverse indicator of self-focused interpersonal styles.

Although the current research did not measure the affective and cognitive components of empathy, when considering the findings in light of previous literature and theoretical explanations of empathy, an interesting pattern unfolds. Preston & De Wall (2003) emphasised the importance of affective empathy in responses towards animals, suggesting individuals who are highly sensitive to emotional signals may be more observant to animal suffering. Consistent with this, agreeableness and prosocialness, both of which are closely tied to emotional warmth, compassion, and sensitivity to others' needs (Baldassari & Grossman 2013; Costa & McCrae, 2008), showed the strongest positive associations with empathy for animals. Whereas a comparatively weaker relationship was observed in conscientiousness, suggesting conscientious

individuals may act responsibly, however, their behaviours may be more guided by duty rather than affective resonance.

The strong negative relationship between empathy for animals and narcissism further supports this interpretation. Narcissistic traits have been associated with reduced affective empathy, even when cognitive perspective-taking remains intact (Ritter et al., 2011). Therefore, with support from previous research it could be argued that this pattern suggests affective sensitivity may play a more prominent role in the relationship between human-animal interactions and social behaviours than expected. Traits closely tied to emotional warmth and resonance showed stronger associations, whereas traits grounded more in duty and reduced emotional attunement showed weaker effects. In turn, affective empathy may function as an underlying mechanism across these findings.

Limitations and Future Directions

Although the findings provided significant support for the hypothesized relationships, they should be considered in light of several limitations.

First, both the AES and the PSA rely on self-report measures of socially desirable characteristics, which may be vulnerable to socially desirable responding and self-perception biases (Edwards & Horst, 1953). It is therefore possible that the strong relationship observed reflects an empathic self-concept or socially desirable responding. This calls for a more extensive approach in future work, for example, social desirability as a control using measures such as the Marlowe Crowne Social Desirability Scale (Crowne & Marlowe, 1960).

Second, the recruited sample also limits the generalizability of the findings. Participants were young adults aged 18–25, mainly recruited through extended social and academic platforms. Therefore, it is possible that the sample partly reflects a similar group in terms of

empathic orientation and personality traits. Future research could benefit from exploring whether the relationships observed in the present study extend to different age groups and populations, providing further insight into how life experiences and generational attitudes influence empathy towards animals, for instance.

Furthermore, the use of the reduced scales may affect comparability with studies that administer the full versions. Where feasible, future research could benefit from using the full scales and replicate the present findings to strengthen comparability and confirm whether the observed relationships remain significant when using the full measures of each construct.

Another consideration concerns the use of multiple regression models. As each model was tested at a significance level of $p < .05$, the probability of error across analyses was increased each time. Future research could benefit from a multivariate approach such as a MANOVA, which allows for multiple variables to be examined simultaneously while controlling the error rate.

Lastly, while the distinction between affective and cognitive empathy, and its role in empathy for animals is theoretically supported by existing literature (Preston & De Waal, 2003), the present study did not directly measure these components. Future research might consider directly measuring both affective and cognitive empathy to examine whether these components differentially explain empathy toward animals and interpersonal tendencies.

Conclusion

The present research aimed to explore the relationship between empathy and concern towards animals and prosocial behaviour, adaptive personality traits and narcissism, questioning whether human-animal interactions extend to broader interpersonal functioning. In doing so, the

study aimed to contribute to the field of research by extending previous literature on empathy, to include empathy for animals and its relationship to social behaviours. Each hypotheses proposed were statistically supported by the findings, suggesting that empathy towards animals meaningfully predicts interpersonal and personality tendencies in young adults. Agreeableness and prosocial behaviours, resembling emotional warmth and sensitivity, showed the strongest relationships to empathy for animals, whereas conscientiousness, a less directly emotional trait, showed a comparatively weaker association. Moreover, narcissism, a trait characterised by empathic deficits and reduced affective resonance, had a strong negative association. Taken together, the current findings suggest that the empathy an individual expresses towards animals may act as a predictive indicator of caregiving behaviours and interpersonal functioning while appearing to more strongly resemble emotional sensitivity and interpersonal warmth.

References

- Allport, G. W. (1937). *Personality: a psychological interpretation*. Psycnet.apa.org.
<https://psycnet.apa.org/record/1938-01964-000>
- Angantyr, M., Eklund, J., & Hansen, E. M. (2011). A Comparison of Empathy for Humans and Empathy for Animals. *Anthrozoös*, 24(4), 369–377.
<https://doi.org/10.2752/175303711x13159027359764>
- Arnett, J. (2000, May 1). *Emerging Adulthood. A Theory of Development From the Late Teens Through the Twenties*. The American Psychologist.
<https://pubmed.ncbi.nlm.nih.gov/10842426/>
- Baldassarri, D., & Grossman, G. (2013). The Effect of Group Attachment and Social Position on Prosocial Behavior. Evidence from Lab-in-the-Field Experiments. *PLoS ONE*, 8(3), e58750. <https://doi.org/10.1371/journal.pone.0058750>
- Baron-Cohen, S., & Wheelwright, S. (2004). The Empathy Quotient: An Investigation of Adults with Asperger Syndrome or High Functioning Autism, and Normal Sex Differences. *Journal of Autism and Developmental Disorders*, 34(2), 163–175.
<https://doi.org/10.1023/b:jadd.0000022607.19833.00>
- Batson, C. D., & Powell, A. A. (2003). Altruism and Prosocial Behavior. *Handbook of Psychology*, 5. <https://doi.org/10.1002/0471264385.wei0519>
- Batson, C. D., Batson, J. G., Slingsby, J. K., Harrell, K. L., Peekna, H. M., & Todd, R. M. (1991). Empathic joy and the empathy-altruism hypothesis. *Journal of Personality and Social Psychology*, 61(3), 413–426. <https://doi.org/10.1037/0022-3514.61.3.413>
- Batson, C. D., Dyck, J. L., Brandt, J. R., Batson, J. G., Powell, A. L., McMaster, M. R., & Griffitt, C. (1988). Five studies testing two new egoistic alternatives to the empathy-altruism hypothesis. *Journal of Personality and Social Psychology*, 55(1), 52–77. <https://doi.org/10.1037/0022-3514.55.1.52>
- Baumeister, R. F., & Leary, M. R. (1995). *APA PsycNet*. Psycnet.apa.org.
<https://psycnet.apa.org/doiLanding?doi=10.1037%2F0033-2909.117.3.497>

- Blair, R. J. R. (2005). Responding to the emotions of others: Dissociating forms of empathy through the study of typical and psychiatric populations. *Consciousness and Cognition*, 14(4), 698–718. <https://doi.org/10.1016/j.concog.2005.06.004>
- Blair, R. J. R. (2010). Empathy, moral development, and aggression: A cognitive neuroscience perspective. In W. F. Arsenio & E. A. Lemerise (Eds.), *Emotions, aggression, and morality in children: Bridging development and psychopathology* (pp. 97–114). American Psychological Association. <https://doi.org/10.1037/12129-005>
- Boyce, C. J., Wood, A. M., & Brown, G. D. A. (2010). The dark side of conscientiousness: Conscientious people experience greater drops in life satisfaction following unemployment. *Journal of Research in Personality*, 44(4), 535–539. <https://doi.org/10.1016/j.jrp.2010.05.001>
- Burgmer, P., Weiss, A., & Ohmann, K. (2019). I don't feel ya: How narcissism shapes empathy. *Self and Identity*, 20(2), 1–17. <https://doi.org/10.1080/15298868.2019.1645730>
- Bzdok, D., Schilbach, L., Vogeley, K., Schneider, K., Laird, A. R., Langner, R., & Eickhoff, S. B. (2012). Parsing the neural correlates of moral cognition: ALE meta-analysis on morality, theory of mind, and empathy. *Brain Structure and Function*, 217(4), 783–796. <https://doi.org/10.1007/s00429-012-0380-y>
- Caprara, G. V., Alessandri, G., Di Giunta, L., Panerai, L., & Eisenberg, N. (2010). The contribution of agreeableness and self-efficacy beliefs to prosociality. *European Journal of Personality*, 24(1), 36–55. <https://doi.org/10.1002/per.739>
- Caprara, G. V., Steca, P., Zelli, A., & Capanna, C. (2005). A New Scale for Measuring Adults' Prosocialness. *European Journal of Psychological Assessment*, 21(2), 77–89.
- Chakrabarti, B., & Baron-Cohen, S. (2006). Empathizing: neurocognitive developmental mechanisms and individual differences. *Understanding Emotions*, 156, 403–417. [https://doi.org/10.1016/s0079-6123\(06\)56022-4](https://doi.org/10.1016/s0079-6123(06)56022-4)
- Chen, N. R. Y., Majeed, N. M., Lai, G. J., Koh, P. S., Kasturiratna, K. T. A. S., Kaur, M., Ho, A. Z. Y., Yong, J. C., & Hartanto, A. (2023). Human–Animal Interaction and Human

- Prosociality: A Meta-Analytic Review of Experimental and Correlational Studies. *Anthrozoös*, 1–20. <https://doi.org/10.1080/08927936.2023.2288745>
- Cialdini, R. B., Schaller, M., Houlihan, D., Arps, K., Fultz, J., & Beaman, A. L. (1987). Empathy-based helping: Is it selflessly or selfishly motivated? *Journal of Personality and Social Psychology*, 52(4), 749–758. <https://doi.org/10.1037/0022-3514.52.4.749>
- Costa, P. T., Jr., & McCrae, R. R. (2008). The Revised NEO Personality Inventory (NEO-PI-R). In G. J. Boyle, G. Matthews, & D. H. Saklofske (Eds.), *The SAGE handbook of personality theory and assessment, Vol. 2. Personality measurement and testing* (pp. 179–198). Sage Publications, Inc. <https://doi.org/10.4135/9781849200479.n9>
- Costa, P., & McCrae, R. R. (1999). *A Five-Factor Theory of Personality*. ResearchGate. https://www.researchgate.net/publication/284978581_A_five-factor_theory_of_personality
- Crowne, D. P., & Marlowe, D. (1960). A new scale of social desirability independent of psychopathology. *Journal of Consulting Psychology*, 24(4), 349–354. <https://doi.org/10.1037/h0047358>
- Cruchinho, P., López-Franco, M. D., Capelas, M. L., Almeida, S., Bennett, P. M., Miranda, M., Teixeira, G., Nunes, E., Lucas, P., & Gaspar, F. (2024). Translation, Cross-Cultural Adaptation, and Validation of Measurement Instruments: A Practical Guideline for Novice Researchers. *Journal of Multidisciplinary Healthcare, Volume 17*, 2701–2728. <https://doi.org/10.2147/jmdh.s419714>
- McCrae, R. R., & Costa, P. T., Jr. (2004). *NEO Five-Factor Inventory--Revised (NEO FFI-R)*[Database record]. APA PsycTests. <https://doi.org/10.1037/t07553-000>
- Eisenberg, N., & Miller, P. A. (1987). The relation of empathy to prosocial and related behaviors. *Psychological Bulletin*, 101(1), 91–119. <https://doi.org/10.1037/0033-2909.101.1.91>

- Roberts, B. W., Lejuez, C., Krueger, R. F., Richards, J. M., & Hill, P. L. (2014). What is conscientiousness and how can it be assessed? *Developmental Psychology*, 50(5), 1315–1330. <https://doi.org/10.1037/a0031109>
- Edwards, A. L., & Horst, P. (1953). Social Desirability as a Variable in 2 Technique Studies. *Educational and Psychological Measurement*, 13(4), 620–625. <https://doi.org/10.1177/001316445301300409>
- Eisenberg, N. (2000). Emotion, Regulation, and Moral Development. *Annual Review of Psychology*, 51(1), 665–697. <https://doi.org/10.1146/annurev.psych.51.1.665>
- Ellingsen, K., Zanella, A. J., Bjerkås, E., & Indrebø, A. (2010). The Relationship between Empathy, Perception of Pain and Attitudes toward Pets among Norwegian Dog Owners. *Anthrozoös*, 23(3), 231–243. <https://doi.org/10.2752/175303710x12750451258931>
- Findlay, L. C., Girardi, A., & Coplan, R. J. (2006). Links between empathy, social behavior, and social understanding in early childhood. *Early Childhood Research Quarterly*, 21(3), 347–359. <https://doi.org/10.1016/j.ecresq.2006.07.009>
- Ghafourifard, M. (2024). Survey fatigue in questionnaire based research: The issues and solutions. *Journal of Caring Sciences*, 13(4). <https://doi.org/10.34172/jcs.33287>
- Gleyzer, R., Felthous, A. R., & Holzer, C. E. (2002). Animal cruelty and psychiatric disorders. *The Journal of the American Academy of Psychiatry and the Law*, 30(2), 257–265. <https://pubmed.ncbi.nlm.nih.gov/12108563/>
- Graziano, W. G., Habashi, M. M., Sheese, B. E., & Tobin, R. M. (2007). Agreeableness, empathy, and helping: A person $\times$ situation perspective. *Journal of Personality and Social Psychology*, 93(4), 583–599. <https://doi.org/10.1037/0022-3514.93.4.583>
- Guillaume Airagnes, du, B., Galam, E., Bunge, L., Hoertel, N., Frédéric Limosin, Philippe Jaury, & Cédric Lemogne. (2021). Personality traits are associated with cognitive empathy in medical students but not with its evolution and interventions to improve it. *Journal of*

- Psychosomatic Research (Print)*, 144, 110410–110410.
<https://doi.org/10.1016/j.jpsychores.2021.110410>
- Hakulinen, C., Elovainio, M., Batty, G. D., Virtanen, M., Kivimäki, M., & Jokela, M. (2015). Personality and alcohol consumption: Pooled analysis of 72,949 adults from eight cohort studies. *Drug and Alcohol Dependence*, 151(1), 110–114.
<https://doi.org/10.1016/j.drugalcdep.2015.03.008>
- Holt-Lunstad, J., Smith, T. B., & Layton, J. B. (2010). Social Relationships and Mortality Risk: a Meta-analytic Review. *PLoS Medicine*, 7(7).
<https://doi.org/10.1371/journal.pmed.1000316>
- Kavanagh, P. S., Signal, T. D., & Taylor, N. (2013). The Dark Triad and animal cruelty: Dark personalities, dark attitudes, and dark behaviors. *Personality and Individual Differences*, 55(6), 666–670. <https://doi.org/10.1016/j.paid.2013.05.019>
- Melchers, M. C., Li, M., Haas, B. W., Reuter, M., Bischoff, L., & Montag, C. (2016). Similar Personality Patterns Are Associated with Empathy in Four Different Countries. *Frontiers in Psychology*, 7(290). <https://doi.org/10.3389/fpsyg.2016.00290>
- Miller, J. D., Back, M. D., Lynam, D. R., & Wright, A. G. C. (2021). Narcissism today: What we know and what we need to learn. *Current Directions in Psychological Science*, 30(6), 519–525. <https://doi.org/10.1177/09637214211044109>
- O’Haire, M. E., McKenzie, S. J., McCune, S., & Slaughter, V. (2014). Effects of Classroom Animal-Assisted Activities on Social Functioning in Children with Autism Spectrum Disorder. *The Journal of Alternative and Complementary Medicine*, 20(3), 162–168.
<https://doi.org/10.1089/acm.2013.0165>
- Paul, E. S. (2000). Empathy with Animals and with Humans: Are They Linked? *Anthrozoös*, 13(4), 194–202. <https://doi.org/10.2752/089279300786999699>
- Preston, S. D., & de Waal, F. B. M. (2002). Empathy: Its ultimate and proximate bases. *Behavioral and Brain Sciences*, 25(1), 1–20.
<https://doi.org/10.1017/s0140525x02000018>

- Raskin, R., & Terry, H. (1988). A principal-components analysis of the Narcissistic Personality Inventory and further evidence of its construct validity. *Journal of Personality and Social Psychology*, 54(5), 890–902. <https://doi.org/10.1037/0022-3514.54.5.890>
- Ritter, K., Dziobek, I., Preißler, S., Rüter, A., Vater, A., Fydrich, T., Lammers, C.-H., Heekeren, H. R., & Roepke, S. (2011). Lack of empathy in patients with narcissistic personality disorder. *Psychiatry Research*, 187(1-2), 241–247. <https://doi.org/10.1016/j.psychres.2010.09.013>
- Robinson, M. D., Irvin, R. L., Hamidreza Fereidouni, & Klein, R. J. (2024). Feelings as a currency of care: A role for agreeableness in emotional reactivity. *Journal of Personality*. <https://doi.org/10.1111/jopy.12951>
- Signal, T. D., & Taylor, N. (2007). Attitude to Animals and Empathy: Comparing Animal Protection and General Community Samples. *Anthrozoös*, 20(2), 125–130. <https://doi.org/10.2752/175303707x207918>
- Tariq, F. T., & Naqvi, D. I. (2020). Relationship between Personality Traits and Prosocial Behavior among Adolescents. *Foundation University Journal of Psychology*, 4(2), 54–63. <https://doi.org/10.33897/fujp.v4i2.79>
- Wells, D. L., & Treacy, K. R. (2024). Pet attachment and owner personality. *Frontiers in Psychiatry*, 15. <https://doi.org/10.3389/fpsy.2024.1406590>
- Wood, L. (2015). The pet factor - companion animals as a conduit for getting to know people, friendship formation and social support. *PLOS ONE*, 10(4), 1–17. <https://doi.org/10.1371/journal.pone.0122085>

Appendix
Appendix A
Recruitment Poster

The poster is titled "How Concern for Animals Relates to Caregiving Behaviours, Personality Traits, and Narcissism in Young Adults". It features a central smartphone displaying the title and an illustration of a man sitting with a dog. The poster is decorated with pink flowers and dashed arrows pointing to four key sections: "About The Study:", "Who can take part?:", "What's Involved?", and "Scan Me!".

DCU Ollscoil Chathair Bhaile Átha Cliath Dublin City University

About The Study:
I am conducting a short online survey exploring how empathy and concern for animals relates to our caregiving behaviors, personality traits and narcissism

Who can take part?:

- ✓ Aged 18-25
- ✓ Can read and understand english
- ✓ Willing to complete an anonymous, short survey

How Concern for Animals Relates to Caregiving Behaviours, Personality Traits, and Narcissism in Young Adults

What's Involved?

- An Anonymous, online survey
- 8 - 10 minutes of your time
- Questions about Personlity, Empathy & Narcissism

Scan Me!

DCU School of Psychology
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Appendix B

Animal Empathy Scale (AES) Survey Items and Psychometric Properties

Item	M	SD	Corrected Item-Total Correlation	α if item deleted
Seeing an Animal in pain upsets me	4.59	.78	.75	.93
I would always try to help if I saw a dog or puppy that seemed to be lost	3.97	.99	.61	.93
Pets have a great influence on my moods.	4.12	1.02	.70	.93
It makes me sad to see an animal on its own in a cage	4.21	.89	.70	.93
Sad films about animals often leave me with a lump in my throat	4.13	.97	.69	.93
I get very angry when I see animals being ill-treated	4.52	.76	.71	.93
It upsets me when I see helpless old animals	4.24	.87	.71	.93
It upsets me to see animals being chased and killed by lions in wildlife programs on TV	3.6	1.11	.58	.93
I find it irritating when dogs try to greet me by jumping up and licking me	3.76	1.12	.72	.93
People often make too much of the feelings and sensitivities of animals	3.59	1.05	.51	.93
I get annoyed by dogs that howl and bark when they are left alone	3.39	1.13	.59	.93
Dogs sometimes whine and whimper for no real reason	3.44	1.11	.62	.93
People who cuddle and kiss their pets in public annoy me	4.07	.97	.71	.93
So long as they're warm and fed, I don't think zoo animals mind being kept in cages	4.02	.98	.67	.93
Many people are over-affectionate towards their pets	3.63	.78	.79	.93
Sometimes I am amazed how upset people get when an old pet dies	4.02	.99	.65	.93

Appendix C

Prosocialness Scale for Adults (PSA) Survey Items and Psychometric Properties

Item Wording	M	SD	Corrected Item-Total Correlation	A if item deleted
I am pleased to help my friends/colleagues in their activities.	4.36	.70	.64	.93
I am empathic with those who are in need.	4.34	.78	.77	.92
I easily put myself in the shoes of those who are in discomfort	3.93	.88	.71	.92
I intensely feel what others feel	3.84	.91	.73	.92
I immediately sense my friends' discomfort even when it is not directly communicated to me	4.06	.88	.65	.93
I try to be close to and take care of those who are in need	3.98	.89	.77	.92
I help immediately those who are in need.	4.00	.91	.75	.92
I do what I can to help others avoid getting in trouble	3.98	.84	.65	.92
I try to console those who are sad	4.18	.80	.75	.93
easily lend money or other things	3.57	1.05	.53	.93
I easily share with friends any good opportunity that comes to me	4.07	.83	.60	.93
I spend time with those friends who feel lonely	3.98	.86	.65	.93
I share the things that I have with my friends	4.15	.84	.72	.92
I am willing to make my knowledge and abilities available to others	4.33	.71	.57	.93

Appendix D

NEO-FFI Agreeableness Survey Items and Psychometric Properties

Item Wording	M	SD	Corrected Item-Total Correlation	A if item deleted
I try to be courteous to everyone I meet.	4.37	.69	.51	.58
I generally try to be thoughtful and considerate	4.42	.68	.50	.59
I would rather cooperate with others than compete with them.	4.02	.89	.49	.57
I believe that most people will take advantage of you if you let them	2.80	1.14	.30	.65
Some people think of me as cold and calculating	3.70	1.02	.43	.59
I'm hard-headed and tough-minded in my attitudes.	2.82	1.06	.22	.68

□

Appendix E

NEO-FFI Conscientiousness Survey Items and Psychometric Properties

Item Wording	M	SD	Corrected Item-Total Correlation	A if item deleted
I strive for excellence in everything I do	4.05	.79	.56	.65
I am a productive person who always gets the job done	3.96	.82	.56	.64
keep my belongings clean and neat.	3.45	1.04	.39	.69
I often get into arguments with my family and co-workers	3.09	.92	.49	.66
If I don't like people, I let them know it	3.22	.87	.32	.72
When I make a commitment, I can always be counted on to follow through.	3.89	.89	.41	.69

Appendix F

Narcissistic Personality Inventory (NPI) Survey Items and Psychometric Properties

Item Wording	M	SD	Corrected Item-Total Correlation	A if item deleted
I know that I am good because everybody keeps telling me so When people compliment me, I sometimes get embarrassed	.20	.40	.52	.66
I am much like everybody else I am an extraordinary person	.73	.45	-.33	.79
I like to be the center of attention I prefer to blend in with the crowd	.26	.44	.49	.66
I am going to be a great person I hope I am going to be successful	.34	.48	.18	.72
I expect a great deal from other people I like to do things for other people	.11	.32	.47	.68
I find it easy to manipulate people I don't like it when I find myself manipulating people	.12	.33	.55	.66
I like having authority over people I don't mind following orders	.29	.45	.38	.68
I insist <u>of</u> getting the respect that is due me I usually get the respect I deserve	.12	.33	.43	.68
I think I am a special person I am no better or nor worse than most people	.23	.42	.51	.66
Everybody likes to hear my stories Sometimes I tell good stories	.15	.36	.44	.68
I am more capable than other people There is a lot that I can learn from other people	.14	.35	.61	.65

Appendix G

Ethical Approval Letter

Scoil na Síceolaíochta
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School of Psychology Ethics Committee (PEC)

Student Name: Sine Quinlan

Project Title: “How Concern for Animals Relates to Caregiving Behaviours, Personality Traits, and Narcissism in Young Adults”

Supervisor: Dr Ciara Mahon

Dear Sine Quinlan,

Thank you for submitting your ethics application.

Based on the information you have provided, your research project has now been approved by the DCU Research Ethics Committee, under delegated authority to the School of Psychology Ethics Committee (PEC). Once in receipt of this notice, you are free to commence your research.

For references purposes, your approval code is: **DCUPEC_2025_135**

Please quote this reference number in all correspondence regarding your project. This reference number should also be included on all participant documentation (e.g. PLS, consent form, debriefing sheet).

You should note that ethics approval has been granted for the specific research project as detailed in your application. Should any amendments be made to the research project, the changes must be approved, in advance. The only exception to this is in cases where there is a small change to participant numbers.

Yours sincerely,

A handwritten signature in black ink, appearing to read 'Maggie Brennan', is shown within a light grey rectangular box.

Dr Maggie Brennan
Deputy Chair, PEC

2nd January, 2026

Note: The PEC complies and retains a full list of all ethics applications approved by it, under delegated authority from REC. This list is then forwarded, on an annual basis, to the REC and, when requested, to the DCU Risk and Compliance Officer.